Risk Assessment

SULZER

Sulzer Middle East - Workshop	**Risk Assessment Team Leader - Prakash Giri**	**Doc Ref. No.: TBA**
Risk Assessment - Milling Machine	**Risk Assessment Team - Prakash Giri**	**Date of Assessment:12/02/2024**

Job Steps Steps in the Activity		**Hazards / Environmental Impacts**			**Perceived Risk**			**Controls**	**Residual Risk**		
		Hazard Description	**Target P.A.E.R.**	**Effects (Consequences)**	**Likelihood**	**Consequences**	**Risk Rating**	**Controls & Recovery Measures Required(Eliminate, Substitute, Engineering Controls, Administrative Controls, PPE)**	**Likelihood**	**Consequences**	**Risk Rating**
1	Machine initial inspection and start up.	*Electricity. *Oil/coolant spillage. *Using damaged tools equipment/machine parts. *Un-authorized persons. *Failure to follow procedure. *Improper guarding. *Failure to inspect all items.	P.R. E.	*Injury to persons. *Damage to property. *Electrocution.	2	3	Medium	*Training and tool box to be provided to the operators how to use the machines. *Do not use damaged tools and equipment. *Pre-use inspection to be conducted by the operatorand maintain the daily checklist. *Any damages or problems identified during the inspection must be rectified before using the machine. *Report any problems identified immediately to your supervisor or the maintenance department. *The electrical cables must be insulated and not damaged. *When filling the coolant tank ensure that it does not overflow and cause spillage. *Oil level must be as per the requirement. If less inform maintenance. *All rotating parts like chucks, open drives, lead screws, etc, shall be properly guarded.	1	3	Low
2	Setting up the job.	*Pinch points. *Loose/Dmaged clamps or jaws. *Oily/ slippery surface of the material. *Unsecured load. *Uneven shape of the material. *Hot surface *Moving and Rotating parts of the machine .	P.E.R.	*Injury to person. *Damage to property. *Fatality/ death.	2	3	Medium	*Persons with long hair must tie their hair and not leave it open. *Check the centre of the gravity of the job before loading it on machine. *Keep clear of dangerous and rotating parts of the machine. *The job must be tightly clamped and must not be loose. *Before releasing the job from the crane it must be properly placed on the machine and must not fall off. *There must be good communication between the departments when tha job is handed ovre for machining. The personnel must provide all the required details to the operator. *Remove loose items like check key, dial gauges etc after setting up the operator. *Use cut resistant gloves for the task and other mandatory PPE.	1	3	Low
3	Machining process.	*Flying objects/chips. *Sharp objects/chips/swarf. *Moving and rotating parts of the machine. *Loose component. *Coolant mist. *Hot chips. *Fumes. *Manual emery on rotating parts. *Overring guards. *Blowing of swarf using compressed air.	P.E.R.	*Injury to person. *Entrapment. *Environmental impact. *Fire *Damage to property *Fatality/death	2	3	Medium	*Training and tool box talks to be conducted. *Machine guards must be used at all times. Do not override any machine guarding. *Do not place your hand on the rotating parts of the machine. *Do not use loose clothing and jewellery while operating the machine. *Carbon dioxide fire extingulsher must be available in accessible location near the machine. *Sanding/ polisshing on rotating parts must not be done by hand. Use the sanding tools provided for polishing. *If the operator has toleave the machine unattended for any reason the machine must be stopped and then the tool post returned to its original position. Do not leave the machine unattended. *Do not touch the swarts and chips or job with bare hands at it might be hot and cause a burn. *Do not clean swart and metal chips using compressed air. Use brush and other tools. *Use proper PPE (coverall, shoes, cut resistant gloves, Etc) mask if matal fumes are present. *Use ear plugs, ear muffs or both when sound level is 85db or above.	1	4	Low
4	General Hazards for all activities	*Manual handling of matterials. *Overweight. *Poor housekeeping. *Poor floor conditions. *Lack of space to manoeuvre. *Poor lighting. *Working alone. *Fatigue. *Distraction. *Exposure to excessive noise. *Slip,trip,fall.	P.E.R.	*Possible back strains and sprains. *Occupational injury. *Chronic health effects. *Personnel injury.	3	4	Medium	*Training and tool box talks on safe work instructions. *The operator mind and eyes must be on the task. Avoid distractions as it might result in an accident. *Continuous housekeeping to be done so that swart and coolant are not on the floor andmachine. *Operators and other team members shall be trained on proper manual handling techniques and they shall be followed. *Proper housekeeping and lean practice. The area must be clean and no dirt, oil or obstructions on the work area. *Materials must be stored in designated storage area and no materials shall be stored on the walkways. *The lighting in the area must be suitable and sufficient for the task to be carried out safely. *Lone working is not permitted, buddy system must be followed. *Sufficient rest to be provided during work and between shifts. *Materials and tools required for the job must be kept nearby so that they are easily accessible to the operetor.	1	4	Low

RA COSHH Control	**Job Title:**	QHSSE Manager	**Approved by:**		**Review Date:**
	Date:		**Signature:**		

(P-People, A- Asset, E- Environment, R- Reputation)